17. Static vs Dynamic Routing Concepts

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Phase: Phase 6 — Routing
Estimated time: 1–2 days
Slug: `17-static-dynamic-routing`

Learning objectives

- Contrast static routes vs OSPF/BGP ideas
- Know when static is enough
- Explain convergence at a high level
- Read simple BGP/OSPF vocabulary

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

Static routes: operator-defined, simple, brittle at scale. Dynamic protocols: OSPF/IS-IS inside orgs; **BGP** between orgs/internet.

```
# You may see learned routes on routers/VMs with routing daemons
ip route show protocol bgp 2>/dev/null
vtysh -c 'show ip route' 2>/dev/null   # FRR/Quagga style labs
```

Lab idea: FRRouting in VMs — not required for every exercise, but valuable later.

Common mistakes

- Injecting static routes that fight BGP
- No documentation of static routes
- Assuming cloud route tables behave like home Wi-Fi

Exercises

Work in `exercises/17-static-dynamic-routing/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/17-static-dynamic-routing/` until after an honest attempt.

Self-check

1. When static routes shine?
2. BGP used where?

3. What is convergence?

Further reading

- https://www.rfc-editor.org/rfc/rfc4271